ASTROBIOLOGY
Volume 3, Number 2, 2003
© Mary Ann Liebert, Inc.

Research Paper

Supercooled Water Brines Within Permafrost—An Unknown Ecological Niche for Microorganisms: A Model for Astrobiology

D. GILICHINSKY,[1] E. RIVKINA,[1] V. SHCHERBAKOVA,[2] K. LAURINAVICHUIS,[2] and J. TIEDJE[3]

Downloaded by Imperial College School Of Med from www.liebertpub.com at 10/05/20. For personal use only.

ABSTRACT

This study describes brine lenses (cryopegs) found in Siberian permafrost derived from ancient marine sediment layers of the Arctic Ocean. The cryopegs were formed and isolated from sediment ~100,000–120,000 years ago. They remain liquid at the *in situ* temperature of −10°C as a result of their high salt content (170–300 g/L). [^{14}C]Glucose is taken up by the cryopeg biomass at −15°C, indicating microbial metabolism at low temperatures in this habitat. Furthermore, aerobic, anaerobic heterotrophs, sulfate reducers, acetogens, and methanogens were detected by most probable number analysis. Two psychrophilic microbes were isolated from the cryopegs, a *Clostridium* and a *Psychrobacter*. The closest relatives of each were previously isolated from Antarctica. The cryopeg econiche might serve as a model for extraterrestrial life, and hence is of particular interest to astrobiology. Key Words: Permafrost—Water brines—Viable microorganisms—Ancient ecosystem—Lake Vostok—Extraterrestrial econiche—Astrobiology—Mars. Astrobiology 3, 331–341.

INTRODUCTION

MICROORGANISMS HAVE SURVIVED in permafrost (Gilichinsky, 2002a) and ice (Abyzov, 1993) at subzero temperatures for hundreds of thousands to millions of years and have been reported to survive at above zero temperatures for dozens to hundreds of millions of years, respectively, in amber (Galippe, 1920; Cano and Borucki, 1995) and saliniferous sediments (Lipman, 1928; Dombrowski, 1963; Vreeland *et al.*, 2000). In all these niches, microorganisms retain viability in the absence of free water. Here we describe a habitat where microorganisms survive in a closed, ancient, aquatic ecosystem within permafrost. These brine lenses were formed during the Quaternary, when the dynamic of polar ocean and geocryological conditions in the high Arctic favored cryopeg formation. This habitat within the permafrost has an age of isolation under subzero conditions on the same order as Lake Vostok, a large Antarctic subglacial lake (Petit *et al.*, 1999).

Because of their distances from the Sun, most

Institutes of [1]Physicochemical and Biological Problems in Soil Science and [2]Biochemistry and Physiology of Microorganisms, Russian Academy of Sciences, Pushchino, Moscow Region, Russia.
[3]Center for Microbial Ecology, Michigan State University, East Lansing, Michigan.

planets of the Solar System are a cryogenic type with the possibility of free water existing only in the presence of high solute content. Subsurface hydrogen on Mars indicates the existence of water (Feldman *et al.*, 2002; Mitrofanov *et al.*, 2002), probably as brine lenses within the planetary permafrost, formed when Mars became dry and cold. These brines, just as their terrestrial models, may contain microorganisms adapted to low temperature and high salinity.

DESCRIPTION OF SALT LENSES

Cryopegs (salt lenses) were exposed in two sites from 10 of 16 closely situated bore holes at Cape Chukochii (70°05′N, 159°55′E) and the Lake Yakutskoe area (69°50′N, 159°30′E) on the tundra zone of the Kolyma River Lowland near the East Siberian Sea coast (Fig. 1). The permafrost here extends up to 800 m deep, and its mean annual temperature varies from −9 to −11°C. The cryopegs are confined to a 20-m-thick marine horizon, sandwiched between nonsaline terrigenous layers at depths of 40–50 m below the surface. Paleontological data show that finely dispersed sand and sandy loams were deposited in shallow lagoons at temperatures slightly above 0°C (Fig. 2, Stage I). After regression of the polar ocean, the water- and methane-saturated bottom sediments were exposed subaerially and froze (Fig. 2, Stage II). Concurrently, because of the pressure caused by freezing, water and gas were released as the freezing front penetrated downward. Then, in the process of cryometamorphism, methane accumulations (Fig. 2, Stage III) and pore methane-hydrates (Rivkina *et al.*, 2001) (Fig. 3) were formed. This was accompanied by a freezing out of salts in the water to form lenses of supercooled sodium chloride brines (Fig. 2, Stage III) with salinities of 170–300 g/L (Table 1). A redox potential indicates an anaerobic environment (Table 2), which is typical for permafrost sediments (Rivkina *et al.*, 1998).

Later (26,000–43,000 years before present), the marine horizon was buried by a 15–20-m-thick

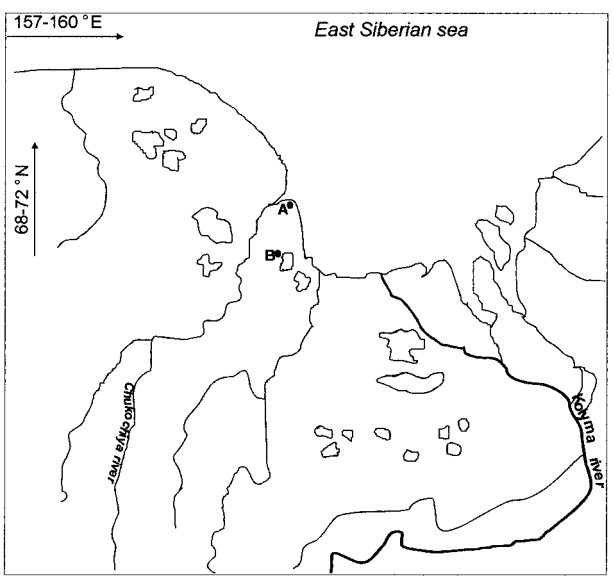

FIG. 1. Schematic map of the study area. Sites with subsurface cryopeg lenses: A, Cape Chukochii; B, Lake Yakutskoe.

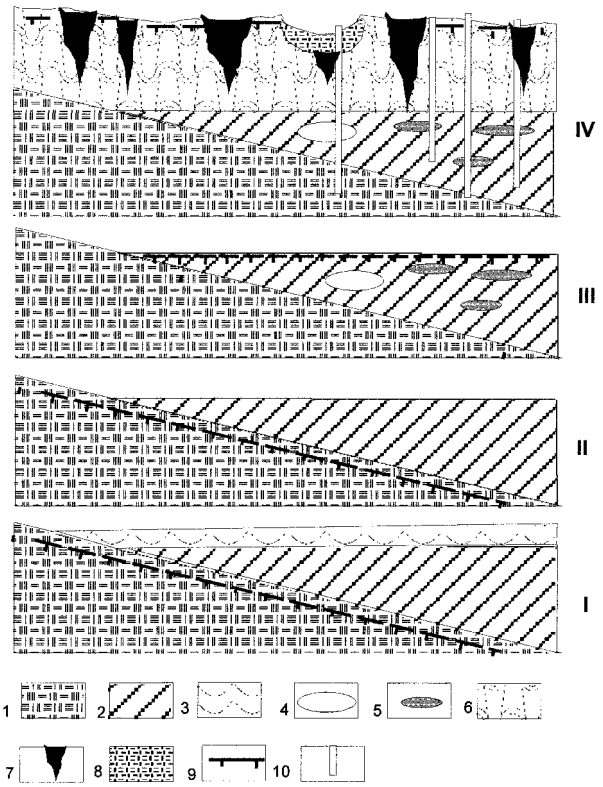

FIG. 2. Stages of cryogenic metamorphism of sediments and subsurface water. Stage I: Submarine conditions. Accumulation of non-frozen marine sediments (QII^{2-3}) (Resolution, 1987). Stage II: Regression of polar ocean (110,000–112,000 years before present) (Romanovskii *et al.*, 1997). Stage III: Subaerial conditions of freezing marine sediments, with formation of frozen saline sediments with gas-hydrates of methane and sandy lenses with cryopegs (QII^4–$QIII^1$). Stage IV: Subaerial conditions of forming late Pleistocene icy complex ($QIII^{2-3}$–QIY). **Insets:** 1, subaerial sediments; 2, marine deposits; 3, marine water; 4, lenses of methane; 5, sandy lenses with cryopegs; 6, icy complex; 7, ice veins; 8, modern sediments; 9, permafrost table; 10, bore holes.

Downloaded by Imperial College School Of Med from www.liebertpub.com at 10/05/20. For personal use only.

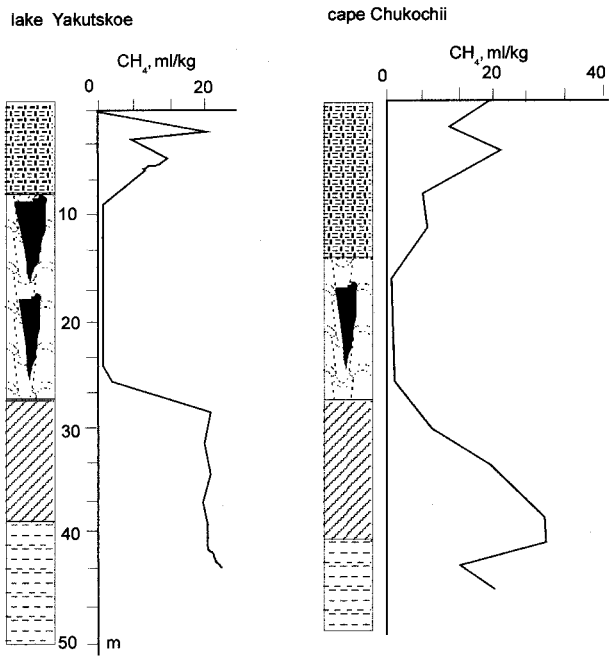

FIG. 3. Distribution of CH_4–hydrates in permafrost sediments near the East Siberian Sea coast (for legend see Fig. 1).

unit of lacustrine-alluvial late-Pleistocene icy complex (Fig. 2, Stage IV) that was syngenetically frozen (concurrently frozen as deposited). This section was built up under harsh climate conditions, as is evident from the pollen patterns. In this time, and as shown by brine salinity (Fotiev, 1997), the mean annual sediment temperatures were lower than present day (−9.5°C at the Lake Yakutskoe area and −10.5°C on Cape Chukochii; Fig. 4c), which correlates with palynological data and characteristics of this period (Sher, 1974). The presence of ice veins in the overlying layer (Figs. 2, Stage IV; and 5) indicates that neither the icy complex nor the brine-containing marine layer has ever been thawed. Within the marine horizon, the lenses occur at different depths, with their thickness varying from 0.5 to 1.5 m and their width from 3 to 5 m. Some of them represent nonartesian water, and some exist under low pressure with a hydrostatic head 2–3 m higher than the enclosed sediments. Different salinities of the brines also confirm their lenticular nature and isolated bedding.

METHODS

Sampling

The drilling system and protocol for the subsequent handling of permafrost cores were designed to ensure recovery of uncontaminated material (Gilichinsky *et al.*, 1989; Shi *et al.*, 1997). However, when the drill enters brine, the latter is contaminated with particles of overlying sediments with their attached microorganisms. Hence, the recovered brine water contained not only indigenous, but also exogenous microflora. This is why one of the goals of the present study was the identification of cryopeg aborigines.

Metabolic activity

The activity of the cryopeg community was determined measuring the uptake of D-[^{14}C]glucose (specific activity, 250 mCi/mol) into the bacterial biomass. Five-milliliter aliquots of brines were supplemented with 2 μCi of glucose and incubated at temperatures from 20 to −15°C for 24 h. To estimate the contribution of contaminating bacteria from the overlying nonmarine sediments, the brine was centrifuged and then filtered to ensure the complete removal of suspended material with adsorbed microorganisms. The re-

TABLE 1. CHEMICAL COMPOSITION OF WATER BRINES

			Concentration [g/L (mol/L)]							
Hole number (depth, m)	E_h *(mV)*	*pH*	*Dry residue*	HCO_3^-	Cl^-	SO_4^{2-}	Ca^{2+}	Mg^{2+}	K^+	Na^+
14/99 (28.0)		7.4	163.03	0.59 (0.01)	99.4 (2.8)	3.84 (0.04)	1.6 (0.04)	8.0 (0.33)	0.8 (0.02)	48.8 (2.12)
15/99 (25.0)	+100	7.3	156.86	0.54 (0.009)	97.3 (2.74)	1.92 (0.02)	1.2 (0.03)	7.7 (0.32)	0.8 (0.02)	47.4 (2.06)
16/99 (11.0)		7.9	120.79	0.37 (0.006)	74.6 (2.1)	1.92 (0.02)	0.8 (0.02)	6.0 (0.25)	0.8 (0.02)	36.3 (1.58)
17/99 (17.0)	+150	7.2	152.65	0.45 (0.007)	93.4 (2.63)	3.4 (0.035)	1.2 (0.03)	7.4 (0.31)	0.8 (0.02)	46.0 (2.0)
15/80 (8.0)		7.5	92.32	0.7 (0.011)	56.1 (1.58)	1.92 (0.02)	2.0 (0.05)	3.8 (0.16)	0.4 (0.01)	27.4 (1.19)
17/80 (16.0)			135.5	0.5 (0.008)	81.3 (2.29)	4.1 (0.043)	1.2 (0.03)	5.5 (0.23)	0.4 (0.01)	42.5 (1.846)

Downloaded by Imperial College School Of Med from www.liebertpub.com at 10/05/20. For personal use only.

TABLE 2. CHEMICAL COMPOSITION OF MARINE AND EMBEDDING SEDIMENTS

Depth (m)	Sediments	pH	Dry residue (%)	Concentration (mmol/100 g)						
				HCO_3^-	Cl^-	SO_4^{2-}	Ca^{2+}	Mg^{2+}	K^+	Na^+
3.5	Modern	7.6	0.04	0.18	0.15	0.36	0.36	0.20	0.07	0.17
5.0		7.8	0.06	0.15	0.20	0.18	0.30	0.20	0.03	0.17
7.5	Icy complex	7.9	0.10	0.70	1.95	0.42	0.54	0.84	0.04	2.30
10.0		7.5	0.24	0.68	1.90	1.4	1.20	1.20	0.03	2.28
15.0		7.2	0.10	0.20	1.12	0.46	1.76	0.16	0.04	0.67
17.5		7.6	0.06	0.22	0.68	0.18	0.40	0.90	0.03	0.34
20.0		7.5	0.09	0.20	0.70	0.36	0.16	0.40	0.04	0.85
22.0	Marine	7.3	0.68	0.65	9.78	0.84	1.40	1.56	0.14	9.80
24.0		7.2	0.66	0.35	9.40	1.02	0.70	0.30	0.27	10.17
26.0		7.1	0.71	0.40	11.31	1.56	0.60	1.16	0.36	11.15
28.0		7.4	0.72	0.28	11.58	0.30	1.64	0.56	0.40	10.80
30.0		7.4	1.14	0.22	15.45	1.72	4.96	1.20	0.32	13.01
32.0		7.3	2.06	0.15	32.20	1.50	7.36	11.90	0.35	23.26
34.0		7.2	0.75	0.10	9.95	1.38	5.16	0.76	0.16	8.02
35.0	Terrigenous	7.2	0.03	0.15	0.25	0.24	0.24	0.30	0.03	0.19
37.0		7.2	0.03	0.15	0.28	0.16	0.20	0.14	0.03	0.30
40.0		7.1	0.04	0.12	0.12	0.42	0.24	0.20	0.03	0.18

moved material amounted to 2 g/100 mL of brine. The autoclaved brine then was inoculated with the equivalent amount of overlying fresh sediments (Fig. 2, inset d), and the metabolic activity was measured by the foregoing method.

Bacterial count

The number of microorganisms in the brine samples was determined by direct counting of cells stained with propidium iodide (Wan *et al.*,

Downloaded by Imperial College School Of Med from www.liebertpub.com at 10/05/20. For personal use only.

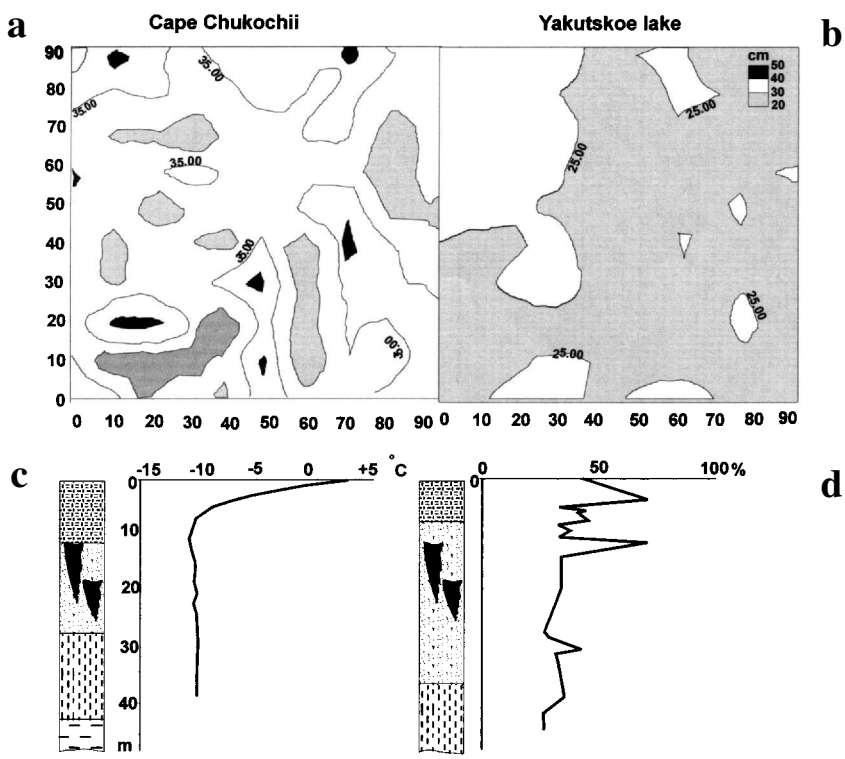

FIG. 4. Characteristics of the layer of annual temperature oscillations: (a and b) the depth of seasonal thawing and (c and d) ground temperature (Hole 4/00) and iciness (Hole 14/99) (for legend see Fig. 1).

FIG. 5. Late Pleistocene icy complex (East Siberian Sea coast, Cape Chukochii).

Downloaded by Imperial College School Of Med from www.liebertpub.com at 10/05/20. For personal use only.

1994) on 0.2-μm-pore-size black polycarbonate membrane filters (Nuclear Corp.). The stained cells were counted using a Lumam fluorescence microscope (Russian) equipped with a blue excitation filter (wavelength 470 nm) and green emission filter (wavelength 530 nm).

Aerobic bacteria were enumerated on R2A (Oxoid) and starch ammonium media using as the inocula frozen sediments or the brine stored in a refrigerator at $-10°C$. The number of anaerobic bacteria of different physiological groups was determined by the five-tube most probable number (MPN) method using the anaerobic technique (Hungate, 1969).

The number of anaerobic heterotrophic bacteria was estimated using a liquid nutrient medium containing (g/L of tap water): KH_2PO_4 (0.7); K_2HPO_4 (0.7); NH_4Cl (0.5); $MgSO_4 \cdot 7H_2O$ (0.1); NaCl (1.0); sodium ascorbate (1.0); glucose (2.0); and peptone (2.0). The number of anaerobic acetogenic microorganisms was estimated using a medium containing (g/L of distilled water): KH_2PO_4 (0.4); NH_4Cl (1.0); NaCl (0.9); $MgCl_2 \cdot 6H_2O$ (0.1); yeast extract (0.1); Tris-HCl (3.0); $Na_2S \cdot 9H_2O$ (0.25); cysteine (0.4); trace element solution (Balch *et al.*, 1979) (1 mL); $NaHCO_3$ (2.0); and resazurin (0.001). Prior to inoculation, the medium was supplemented with 1 m*M* sodium molybdate (an inhibitor of sulfate reduction) and 10 m*M* 2-bromoethanesulfonic acid (an inhibitor of methanogenesis). The growth substrate was a mixture of 80% H_2 and 20% CO_2.

The number of methanogenic bacteria was estimated using mineral medium 2 (Balch *et al.*, 1979) and sodium acetate (2 g/L), trimethylamine (1 g/L), or a mixture of H_2 and CO_2 as the growth substrates. The growth of heterotrophic, acetogenic, and methanogenic bacteria was evaluated from the amount of metabolic products (methane and fatty acids), which were determined chromatographically (Kotelnikova *et al.*, 1998).

The number of sulfate-reducing bacteria was estimated from the amount of hydrogen sulfide (Cline, 1969) produced in the liquid medium of Widdel and Pfennig (1982) supplemented with sodium lactate (1.5 g/L).

Halophilic microorganisms of the respective physiological groups were enumerated in the same media but supplemented with 100 g/L NaCl. The inoculated test tubes were incubated at 5 and 18°C for 3 months.

Downloaded by Imperial College School Of Med from www.liebertpub.com at 10/05/20. For personal use only.

Enrichment and isolation

The enrichment culture yielding strain 14D1 was obtained by incubating a mixture of the medium for heterotrophic bacteria with 10% brine at 5°C for 3 months in the dark until the culture turbidity (optical density) at 600 nm reached a value of 0.6. The colonies of the enriched culture were obtained on a thin layer of 2% Difco agar in anaerobic Bellco glass tubes. The colonies of strains 1pS and 2pS were grown on agar plates containing (g/L): Na_2HPO_4 (11.2); KH_2PO_4 (4.0); NH_4Cl (2.0); NaCl (4.0); $MgCl_2 \cdot 6H_2O$ (0.1); $CaCl_2$ (0.01); $FeSO_4 \cdot 7H_2O$ (0.005); trace element solution SL-10 (Widdel and Pfennig, 1982) (10 mL); sodium acetate (10.0); and agar (20).

Microscopic examination

Cell morphology was examined using an Opton ICM 405 phase-contrast microscope at a magnification of ×350.

Phenotypic tests

Gram stain, the presence of oxidase and catalase, and glucose fermentation were determined by standard methods (Smibert and Krieg, 1994).

DNA isolation

DNA was isolated by the method of Marmur (1961) and analyzed for nucleotide composition by the method of thermal denaturation using a Pye Unicam SP1800 spectrophotometer.

Phylogenetic analysis

The phylogenetic relationship between strains 14D1 and 2pS and the reference strains was studied using 16S rRNA gene sequences by distance matrix, parsimony, and maximum likelihood analyses. Additional sequences were retrieved from GenBank by BLAST searching. For comparison of cryopeg 16S rDNA sequences the NCBI and RDP databases were used. The nucleotide sequences of strain 14D1 and 2pS have been deposited in the GenBank database under the respective accession numbers AY117755 and AF5177557.

RESULTS

As expected with decreasing incubation temperature the incorporation of ^{14}C from glucose by the cryopeg microbial biomass decreased from 200,000 cpm at room temperature but remained significant even at −15°C (8,500 cpm). The metabolic activity measured in the autoclaved brine inoculated with the drilling contaminants was extremely low in comparison with the metabolic activity of native brine (Fig. 6). Consequently, the exogenous microflora that spilled contamination into the brine due to drilling had minimal metabolic activity.

The total number of microorganisms found in cryopegs, 10^7 cells/mL (Table 2), is close to that of common aquatic habitats and frozen sediments (Vorobyova *et al.*, 1997). The population of culturable aerobic bacteria in the brine, up to 80–95% and represented by gram-positive cocci and coccobacilli, varied from hundreds to hundreds of thousand per milliliter of water (Table 3), and in marine deposits, for comparison, 10^2–10^3 cells/g (Table 4). The optimal growing temperature for cryopeg microorganisms was usually 4–5°C (Table 3), in contrast to bacteria from the embedding sediments where the optimum growth was at room temperature (Table 4).

Two pure cultures of aerobic strains of the same species 1pS and 2pS were obtained by incubating brine enrichments on acetate-containing agar media at 18°C (Fig. 7b). Strains 1pS and 2pS were oxidase- and catalase-positive cocci and coccobacilli (1.5 × 3 μm), respectively, and were able to grow at subzero temperatures and at NaCl con-

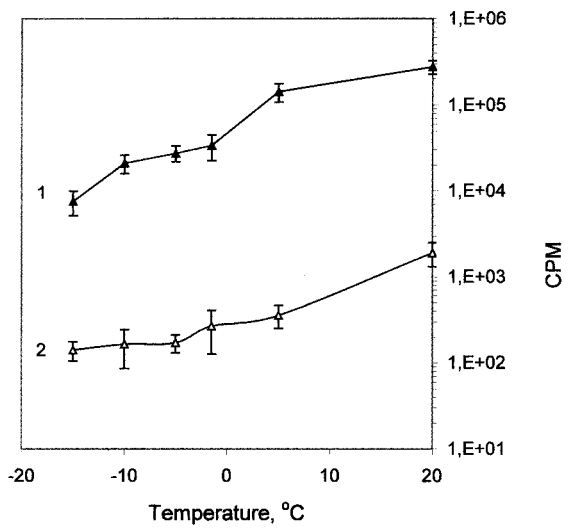

FIG. 6. Incorporation of D-[U-^{14}C]glucose by native bacterial population in cryopeg (1) and bacteria from overlying nonsaline sediments added in autoclaved brine (2).

TABLE 3. MPN ESTIMATES OF MICROORGANISMS IN BRINES, COMPARED WITH A DIRECT COUNT OF 10^7 CELLS/mL

	Viable microorganisms (cells/mL)							
	Aerobic bacteria				*Anaerobic bacteria*			
	Hole number							
Incubation temperature (°C)	*14/99*	*15/99*	*16/99*	*17/99*	*Heterotrophs*	*Sulfate reducers*	*Acetogens*	*Methanogens*
−1.5	5×10^4	9×10^2	2×10^4	0	Not determined			
4–5	6×10^5	6×10^4	5×10^5	1×10^2	2.0×10^2	2.0×10^5	1.0×10^1	1.0×10^2
18–20	2×10^6	8×10^3	7×10^4	7×10^1	2.0×10^1	2.0×10^2	0	0
5[a]	Not determined				0	2.0×10^6	0	0
18[a]					0	0	0	0

[a]Incubation with basal media contained 100 g/L NaCl.

centrations varying from 0% to 6.0% (optimum 0.4–0.8%). They are halotolerant, as was the case for other permafrost isolates (Vishnivetskaya *et al.*, 2000).

Strain 2pS is a close relative (97% similarity) of *Psychrobacter glacincola* isolated from Antarctic sea ice (Bowman *et al.*, 1997) (Fig. 8). The G+C contents of the DNA of strains 1pS and 2pS were close (46.4 and 46.0 mol %, respectively), suggesting that they belong to the same species.

Strain 2pS grew at temperatures of −5 to +35°C with an optimum growth at +16/18°C. This indicates that this strain is psychrotrophic, but with psychrophilic properties that are typical of the permafrost microbial community (Gilichinsky, 2002a). For instance, the strain of the genus *Psychrobacter* isolated from the overlying late Pleistocene sediments has the same temperature range for growth (Ponder *et al.*, 2003).

Anaerobic groups of heterotrophs, methanogens, sulfate reducers, and acetogens were detected; their numbers varied from tens (acetogens) to 2×10^6 cells/mL (halophilic sulfate reducers). The number of heterotrophs was maximal (2×10^2 cells/mL) in a mineral medium incubated at 5°C, whereas their number at 18°C was an order of magnitude lower (2×10^1 cells/mL). The largest number of methanogens (10^2 cells/mL) was detected in a medium with acetate. Halophilic acetogens, het-

TABLE 4. NUMBER OF VIABLE MICROORGANISMS IN PERMANENTLY FROZEN MARINE SEDIMENTS EMBEDDING CRYOPEGS (HOLE 14/99)

	Concentration (cell/g) in							
	Medium R2A				*Starch-ammonium agar*			
	At incubation temperature (°C)							
Depth (m)	*30*	*20*	*4*	*0*	*30*	*20*	*4*	*0*
21.60–21.70	82	4,412	577	330	0	4,907	82	41
23.40–23.50	322	689	506	0	46	322	0	0
24.15–24.20	87	0	2,687	0	0	130	43	0
25.42–25.46	0	250	125	0	42	167	167	0
26.53–26.58	367	128	37	0	37	0	0	0
27.35–27.40	342	3,584	85	0	85	256	0	0
28.45–28.50	231	3,892	501	270	77	231	193	39
29.10–29.17	198	266	238	40	40	158	40	0
31.85–31.90	1,915	849	266	0	162	ND	0	0
34.40–34.45	479	468	205	0	0	131	44	44
35.80–35.90	205	28	205	0	0	0	121	0
36.70–36.75	84	867	28	0	180	85	1,708	0
37.85–37.90	17	1,0958	122	17	424	121	152	61

ND, not determined.

Downloaded by Imperial College School Of Med from www.liebertpub.com at 10/05/20. For personal use only.

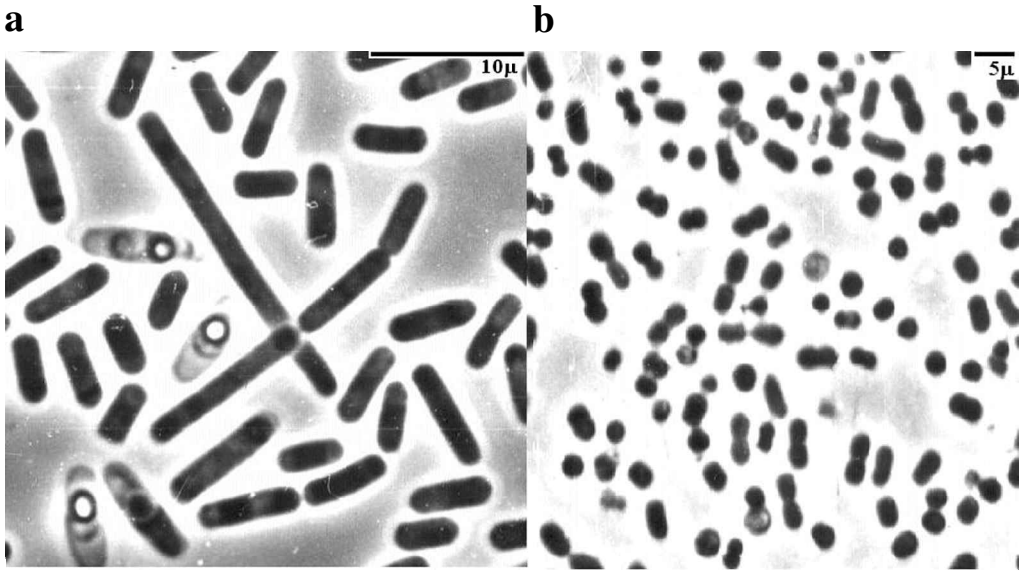

FIG. 7. Phase-contrast microphotograph of *Clostridium* sp., strain 14D1 (a) and *Psychrobacter* sp., strain 2pS (b) isolated from permafrost water brine.

erotrophs, and methanogens were not detected (Table 2).

Under inoculation of brine water with heterotrophic media and culturing in the tubes with rubber stoppers in a N_2 atmosphere at 5°C, spore-forming rod-shaped cells were the dominant forms of the psychrophile community present. A pure culture of strain 14D1 was obtained by successive transfer of single colonies to glucose-containing liquid medium. These cells were somewhat polymorphic, gram-positive, motile rods (1.5×7 μm) with a central endospore (Fig. 7a). The strain did not contain catalase and oxidase and are strictly anaerobic. The nucleotide sequence of 16S rRNA gene of strain 14D1 is closely related to that of a species of *Clostridium* isolated from Antarctic sea ice (Spielmeyer *et al.*, 1993) and Antarctic bacterial mats (Spring *et al.*, 2003) (Fig. 9). The 16S rRNA gene sequence of strain 14D1 shows 98% similarity with *Clostridium frigoris*, *Clostridium bowmanii*, *Clostridium lacusfryxellense*, and *Clostridium estertheticum*. Strain 14D1 can grow at temperatures from 18 to −5°C, with an optimum growth and activity at +5–6°C, suggesting that the strain is a strict psychrophile (Morita, 1975). At transition in culturing temperatures from positive to subzero we observed an increase of both the optimal and the acceptable growth concentrations of NaCl. The optimal concentration of NaCl for the growth of strain 14D1 at 5°C was 0.5%. At −5°C the strain was able to grow at NaCl concentrations varying from 0% to 10.0%, with optimal growth at 1% NaCl (Fig. 10).

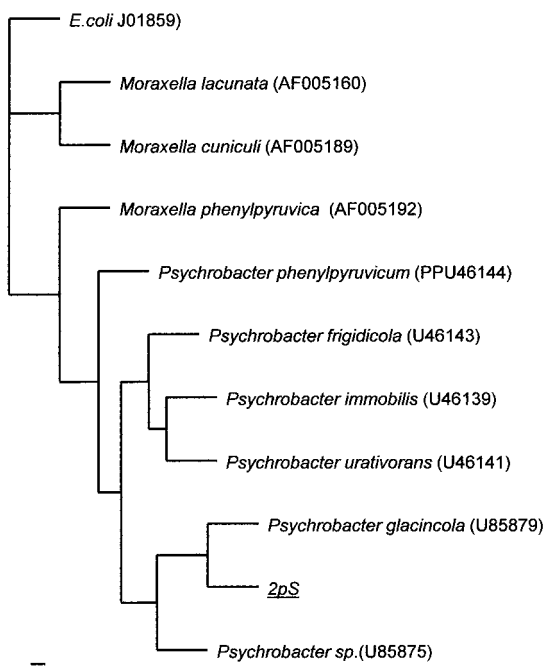

FIG. 8. Phylogenetic dendrogram based on the 16S rDNA sequence showing the position of strain 2pS among representatives of the genus *Psychrobacter* and *Moraxella*. Outgroup sequence used was *Escherichia coli*. The scale bar indicates 10 substitutions per 100 nucleotides. GenBank accession numbers for these bacteria are given in parentheses following the species names.

DISCUSSION

At Pleistocene and modern ground temperatures, fine-grained sediments are firmly bound

Downloaded by Imperial College School Of Med from www.liebertpub.com at 10/05/20. For personal use only.

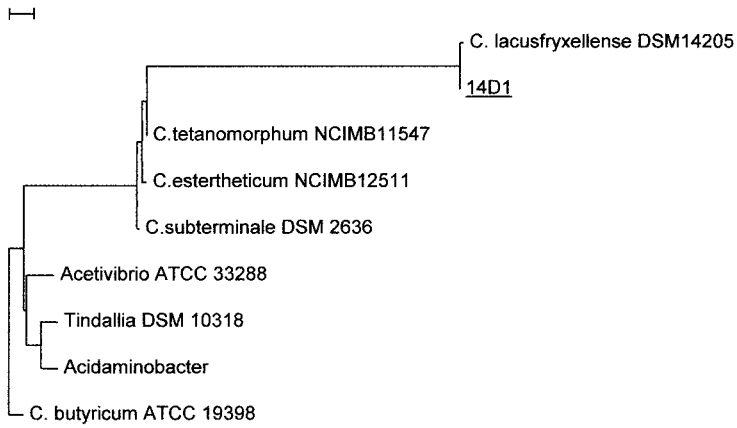

FIG. 9. Phylogenetic dendrogram based on the 16S rDNA sequence showing the position of strain 14D1 among representatives of the genus *Clostridium*. Outgroup sequence used was *Clostridium butyricum*. The scale bar indicates 10 substitutions per 100 nucleotides. GenBank accession numbers for these bacteria are given in parentheses following the species names.

Downloaded by Imperial College School Of Med from www.liebertpub.com at 10/05/20. For personal use only.

together by ice (excess ice volumes of 20–60%; Fig. 4d). Therefore, below the seasonally thawing layer, which is <0.5 m (Fig. 4a and b), infiltration into the cryopeg lenses is precluded. Because of the high ice content, the pores are completely filled with ice, excluding any migration within the stratum. Although unfrozen water films in icy complex loams vary from 3% to 5%, we are convinced that the thermal diffusion of cells within these films is impossible. First, the cells are adsorbed on the surface of mineral particles, and second, their typical size (0.3–0.5 μm) is not comparable with the size of the thickness of these films [10^{-3} μm (Andersen, 1967)]. Also, the cells found there, which vary in size from 1.5 to 7 μm (Fig. 7), could not get into the brine. This is why one can conclude that, in general, the cryopeg microbial community was inherited from the marine sediments.

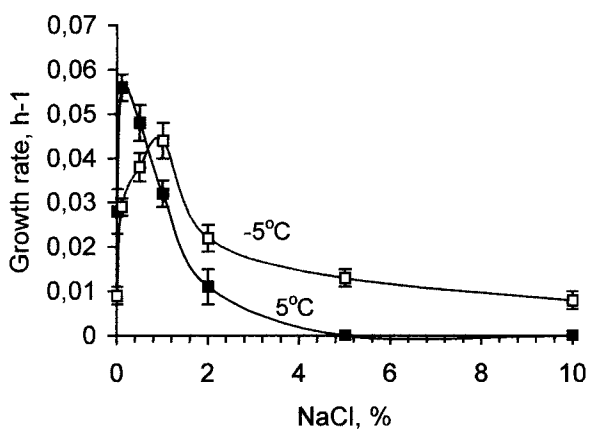

FIG. 10. Shift of strain 14D1 optimum salinity for growth at +5 and −5°C.

The previous study showed that permafrost does not contain psychrophilic bacteria (Vorobyova *et al.*, 1997). Conversely, the bacteria isolated from cryopegs were psychrophilic, which is typical for cold water versus a frozen soil environment (Gounot, 1986). The two isolates may be members of the same trophic chain; anaerobic strain 14D1 produces lactate and butyrate, while the aerobic strain 2pS utilizes these substrates as the sources of carbon and energy. Hence, they may be cryopeg indigenous species adapted to conditions of high salinity. For example, the salinity optimum for growth of strain 14D1 increased with a decrease in subzero temperatures (Fig. 10).

Communities such as these might be found in Lake Vostok, since both ecosystems became isolated in the mid Pleistocene (Petit *et al.*, 1999) and under supercooled conditions. However, Lake Vostok may have less salt and fewer energy resources. At Lake Vostok, concern over contamination of the lake by entry of the drill and the associated well fluids into the water has presently halted the drilling until a means is devised to prevent water contamination. In the meantime, studies of the microbiota and ecosystem of the cryopegs, both of which are less costly and with less concern for contamination, could advance knowledge on such systems.

The ability of microorganisms to live in a variety of natural environments continually forces us to redefine the limits of both life and the biosphere. Biotic survival in the aquatic environment

on a geological time scale indicates unknown bacterial adaptations. It remains to be determined whether the salt tolerance of these cells may also be associated with cold tolerance. Experiments showed that halophilic bacterial remain viable at 80°C in the presence of 25% NaCl (Mancinelli and Landheim, 2002).

A characteristic property of a microbial community of cryopegs is the presence of halophilic microorganisms that had never been isolated from permafrost sediments. Sulfate reducers detected in cryopegs are halophilic and psychrophilic at the same time.

The long-term survival of bacteria, both aerobic and anaerobic, in permafrost sediments, based on the detection of several metabolic reactions, suggests that these microbes can cope with these stresses, which include repair of their DNA damage resulting from background soil radiation, repair of cell membranes, and other vital functions to maintain the cell's viability (Rivkina *et al*., 2000, 2003). The microbial activity detected in cryopegs at temperatures as low as −15°C also documents the fact that subzero temperatures themselves do not exclude biochemical reactions. The observed consumption of labeled glucose at these temperatures and salinity provides reason to conclude that in supercooled water the metabolic strategy of microbial survival operates and that this strategy does not accept that cells can multiply *in situ*.

CONCLUSION

The lenses of brines within permafrost (cryopegs) make up the only habitat on Earth that is characterized by permanently subzero temperatures, high salinity, and isolation from the influence of external factors during geological time. From the astrobiological perspective, mineral-enriched brines provide the only opportunity for free water within the Martian subsurface permafrost (Gilichinsky, 2002b). Arctic cryopegs, preserved dozens to hundreds of thousands of years and more, represent the terrestrial model of such exobiological niches with their unique halotolerant and aerobic and anaerobic psychrophilic community, a plausible prototype for Martian microbial life. Of course, one cannot make a generalized statement based on one or two individual isolates, and further investigations are warranted.

ACKNOWLEDGMENTS

This study was supported by the Russian Foundation for Basic Research (Projects 01-04-49084 and 01-05-65043) and by the NASA Astrobiology Institute. The radiocarbon dating was done by Beta Analytic Radiocarbon Dating Laboratory. We thank Dr. A. Tsapin (Jet Propulsion Laboratory) for help with 16S rRNA gene sequencing of strain 2pS and Dr. L. Petrovskaya (Institute of Biochemistry, Russian Academy of Sciences) for similar help with sequencing of strain 14D1. The depths of seasonally thawing ground were observed under the framework of the of U.S. NSF Circumpolar Active Layer Monitoring (CALM) Program. The authors thank Dr. Jerry Brown (International Permafrost Association) for his help in the manuscript's preparation and attention to Permafrost Astrobiology.

REFERENCES

Abyzov, S. (1993) The microorganisms in the Antarctic ice. In *Antarctic Microbiology*, edited by E.I. Friedmann, Wiley, New York, pp. 265–297.

Andersen, D. (1967) Ice nucleation and the substrate-ice interface. *Nature* 216, 563–566.

Balch, W., Fox, G., Magrum, L., and Wolfe, R. (1979) Methanogens: reevalution of a unique biological group. *Microbiol. Rev*. 43, 260–296.

Bowman, J., McCammon, S., Brown, M., Nichols, D., and McMeekin, T. (1997) Diversity and association of psychrophilic bacteria in Antarctic sea ice. *Appl. Environ. Microbiol.* 63, 3068–3078.

Cano, R. and Borucki, M. (1995) Revival and identification of bacterial spores in 25- to 40-million-year-old Dominican amber. *Science* 268, 1060–1064.

Cline, J. (1969) Spectrophotometric determination of hydrogen sulfide in natural waters. *Limnol. Oceanogr*. 14, 444–458.

Dombrowski, H. (1963) Bacteria from paleozoic salt deposits. *Ann. N.Y. Acad. Sci*. 108, 453–460.

Feldman, W., Boynton, W., Tokar, R., Prettyman, T., Gasnault, O., Squyres, S., Elphic, R., Lawrence, D., Lawson, S., Maurice, S., McKinney, G., Moore, K., and Reedy, R. (2002) Global distribution of Neutrons from Mars: results from Mars Odyssey. *Science* 297, 75–78.

Fotiev, S.M. (1997) Hydrochemical method for estimation of paleotemperature of rocks on the Arctic coast [in Russian]. *Earth Cryosphere* 1(2), 29–35.

Galippe, V. (1920) Recherches sur la resistance de microzymes a laction du temps et sur leur survivance dans l'amber. *C. R. Acad. Sci*. 170, 856–858.

Gilichinsky, D., Khlebnikova, G., Zvyagintsev, D., Fyodorov-Davydov, D., and Kudryavtseva, N. (1989) Microbiology of sedimentary materials in the permafrost zone. *Int. Geol. Rev*. 31, 847–858.

Downloaded by Imperial College School Of Med from www.liebertpub.com at 10/05/20. For personal use only.

Gilichinsky, D. (2002a) Permafrost as a microbial habitat. In *Encyclopedia of Environmental Microbiology*, edited by G. Bitton, Wiley, New York, pp. 932–956.

Gilichinsky, D. (2002b) Permafrost model of extraterrestrial habitat. In *Astrobiology*, edited by G. Horneck and C. Baumstark-Khan, Springer-Verlag, Berlin, pp. 125–142.

Gounot, A.-M. (1986) Psychrophilic and psychrotrophic microorganisms. *Experientia* 42, 1192–1197.

Hungate, R.E. (1969) A roll tube method for cultivation of strict anaerobes. In *Methods in Microbiology*, edited by J. Norris and D. Ribbons, Academic Press, New York, pp. 117–132.

Kotelnikova, S., Macario, J., and Pedersen, K. (1998) *Methanobacterium subterraneum* sp. nov., a new alkaliphilic, eurythermic and halotolerant methanogen isolated from deep granitic groundwater. *Int. J. Syst. Bacteriol.* 48, 357–367.

Lipman, C. (1928) The discovery of living microorganisms in ancient sediments. *Science* 68, 272–273.

Mancinelli, R. and Landheim, R. (2002) Mars, permafrost and halophiles [NAI publication 76]. In *International Workshop on Water in the Upper Martian Surface, Abstracts*, Potsdam, Germany, Abstract #6.

Marmur, J. (1961) A procedure for the isolation DNA from microorganisms. *J. Mol. Biol.* 3, 208–218.

Mitrofanov, I., Anfimov, D., Kozyrev, A., Litvak, M., Sanin, A., Tret'yakov, V., Krylov, A., Shvetsov, V., Boynton, W., Shinohara, C., Hamara, D., and Saunders, R. (2002) Maps of subsurface hydrogen from the high energy neutron detector, Mars Odyssey. *Science* 297, 78–81.

Morita, R. (1975) Psychrophilic bacteria. *Bacteriol. Rev.* 39, 144–167.

Petit, J., Jouzel, J., Raynaud, D., Barkov, N., Barnola, J.-M., Basile, I., Bender, M., Chappellaz, J., Davis, M., Delaygue, G., Delmotte, M., Kotlyakov, V., Legrand, M., Lipenkov, V. , Lorius, C., Pépin, L., Ritz, C., Saltzman, E., and Stievenard, M. (1999) Climate and atmospheric history of the past 420.000 years from the Vostok ice core, Antarctica. *Nature* 399, 429–436.

Ponder, M., Bergholz, P., Mindock, C., Hollingsworth, R., Tiedje, J., and Tomashow, M. (2003) Growth and metabolic activity of ancient permafrost bacteria in cold, low water activity conditions [abstract]. *Astrobiology* 2, 543.

Resolutions of Interdepartmental Conference on the Quaternary System of the East of USSR. Explanationary notes to the regional stratigraphic charts of the Quaternary deposits [in Russian]. (1987) *Magadan* 1–241.

Rivkina, E., Gilichinsky, D., Wagener, S., Tiedje, J., and McGrath, J. (1998) Biogeochemical activity of anaerobic microorganisms from buried permafrost sediments. *Geomicrobiol. J.* 15, 187–193.

Rivkina, E., Friedmann, E.I., McKay, C., and Gilichinsky, D. (2000) Metabolic activity of permafrost bacteria below the freezing point. *Appl. Environ. Microbiol.* 66, 3230–3233.

Rivkina, E., Gilichinsky, D., McKay, C., and Dallimore, S. (2001) Methane distribution in permafrost: evidence for an interpore pressure methane hydrate. In *NATO Series*: *Permafrost Response on Economic Development, Environmental Security and Natural Potential*, edited by R. Paepe and V. Melnikov, Kluwer Academic Publishers, Dordrecht, The Netherlands, pp. 487–496.

Rivkina, E., Laurinavichius, K., McGrath, J., Tiedje, J., Shcherbakova, V., and Gilichinsky, D. (2003) Microbial life in permafrost. *Adv. Space Res.* (in press).

Sher, A.V. (1974) Pleistocene mammals and stratigraphy of the Far Northeast USSR and North America. *Int. Geol. Rev.* 16, 1–89.

Shi, T., Reevs, R., Gilichinsky, D., and Friedmann, E.I. (1997) Characterization of viable bacteria from Siberian permafrost by 16S rDNA sequencing. *Microb. Ecol.* 33, 169–179.

Smibert, R. and Krieg, N. (1994) Phenotypic characterization. In *Methods for General and Molecular Bacteriology*, edited by P. Gerhardt, R. Murrey, W. Wood, and N. Krieg, American Society for Microbiology, Washington, DC, pp. 607–654.

Spielmeyer, W., McMeekin, T., Miller, J., and Franzmann, P. (1993) Phylogeny of the Antarctic bacterium, *Carnobacterium alterfunditum. Polar Biol.* 13, 501–503.

Spring, S., Merkhoffer, B., Weiss, N., Kroppetstedt, R., Hippe, H., and Stackebrandt, E. (2003) Characterization of novel psychrophilic clostridia from an Antarctic microbial mat: description of *Clostridium frigoris* sp. nov., *C. lacusfryxellense* sp. nov., *C. bowmanii* sp. nov. and *C. psychrophilum* sp. nov. and reclassification of *C. laramiense* as *C. estertheticum* subsp. *laramiense* subsp. nov. *Int. J. Syst. Evol. Microbiol.* (in press).

Vishnivetskaya, T., Kathariou, S., McGrath, J., Gilichinsky, D., and Tiedje, J. (2000) Low-temperature recovery strategies for the isolation of bacteria from ancient permafrost sediments. *Extremophiles* 3, 165–173.

Vorobyova, E., Soina, V., Gorlenko, M., Minkovskaya, N., Mamukelashvili, A., Zalinova, N., Gilichinsky, D., Rivkina, E., and Vishnivetskaya, T. (1997) The deep cold biosphere: facts and hypothesis. *FEMS Microbiol. Rev.* 20, 277–290.

Vreeland, R., Rosenzweig, W., and Powers, D. (2000) Isolation of a 250 million-year-old halotolerant bacterium from a primary salt crystal. *Nature* 407, 897–900.

Wan, C., Sigh, R., and Lau, B. (1994) A simple fluorometric assay for the determination of cell numbers. *J. Immunol. Methods* 173, 265–272.

Widdel, F. and Pfennig, N. (1982) Studies on dissimilatory sulfate-reducing bacteria that decompose fatty acids. Incomplete oxidation of propionate by *Desulfobulbus propionicus* gen. sp. nov. *Arch. Microbiol.* 131, 360–365.

Address reprint requests to:
Dr. D. Gilichinsky
Institute of Physicochemical and Biological Problems in Soil Science
Russian Academy of Sciences
142290, Pushchino, Moscow Region
Russia

E-mail: gilichin@issp.serpukhov.su

Downloaded by Imperial College School Of Med from www.liebertpub.com at 10/05/20. For personal use only.